Decomposition in DBMS | Lossless | Lossy

Decomposition of a Relation-

> The process of breaking up or dividing a single relation into two or more sub-relations is called as decomposition of a relation.

Properties of Decomposition-

The following two properties must be followed when decomposing a given relation-

1. Lossless decomposition-

Lossless decomposition ensures-

- No information is lost from the original relation during decomposition.
- When the sub-relations are joined back, the same relation is obtained that was decomposed.
- Every decomposition must always be lossless.

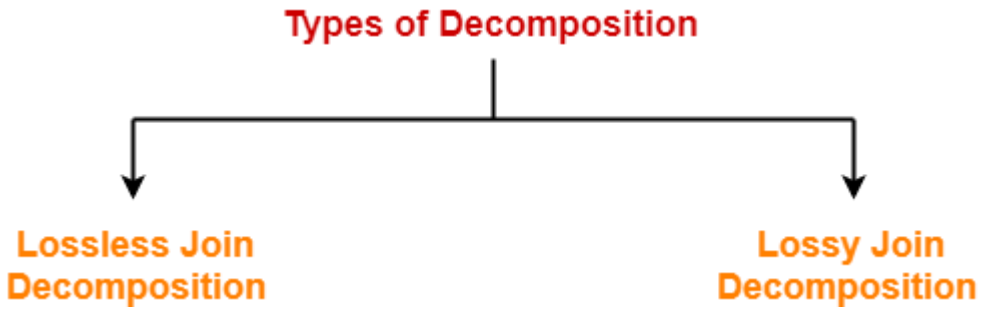
2. Dependency Preservation-

Dependency preservation ensures-

- None of the functional dependencies that holds on the original relation are lost.
- The sub relations still hold or satisfy the functional dependencies of the original relation.

Types of Decomposition-

Decomposition of a relation can be completed in the following two ways-



1. Lossless Join Decomposition-

- Consider there is a relation R which is decomposed into sub relations R_1 , R_2 , …. , R_n.
- This decomposition is called lossless join decomposition when the join of the sub relations results in the same relation R that was decomposed.
- For lossless join decomposition, we always have-

$$R_1 \bowtie R_2 \bowtie R_3 \ldots\ldots \bowtie R_n = R$$

where ⋈ is a natural join operator

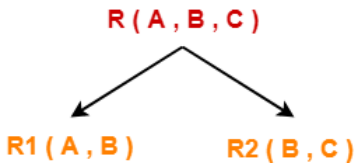
Example-

Consider the following relation R(A , B , C)-

A	B	C
1	2	1
2	5	3
3	3	3

R(A , B , C)

Consider this relation is decomposed into two sub relations R_1(A , B) and R_2(B , C)-



The two sub relations are-

A	B
1	2
2	5
3	3

R_1(A , B)

B	C
2	1
5	3
3	3

R_2(B , C)

Now, let us check whether this decomposition is lossless or not.

For lossless decomposition, we must have-

$$R_1 \bowtie R_2 = R$$

Now, if we perform the natural join ($\bowtie$) of the sub relations R_1 and R_2 , we get-

A	B	C
1	2	1
2	5	3
3	3	3

This relation is same as the original relation R.

Thus, we conclude that the above decomposition is lossless join decomposition.

NOTE-

- Lossless join decomposition is also known as **non-additive join decomposition.**
- This is because the resultant relation after joining the sub relations is same as the decomposed relation.
- No extraneous tuples appear after joining of the sub-relations.

2. Lossy Join Decomposition-

- Consider there is a relation R which is decomposed into sub relations R_1, R_2 , …., R_n.
- This decomposition is called lossy join decomposition when the join of the sub-relations does not result in the same relation R that was decomposed.
- The natural join of the sub-relations is always found to have some extraneous tuples.
- For lossy join decomposition, we always have-

$$R_1 \bowtie R_2 \bowtie R_3 \ldots\ldots\ldots \bowtie R_n \supset R$$

where ⋈ is a natural join operator

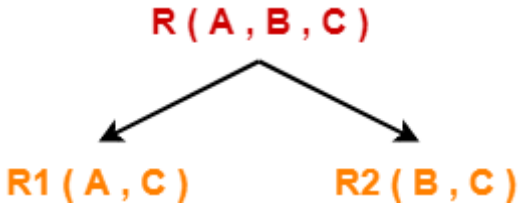
Example-

Consider the following relation R(A , B , C)-

A	B	C
1	2	1
2	5	3
3	3	3

R(A , B , C)

Consider this relation is decomposed into two sub relations as R_1(A , C) and R_2(B , C)-



The two sub relations are-

A	C
1	1
2	3
3	3

R_1(A , B)

B	C
2	1
5	3
3	3

R_2(B , C)

Now, let us check whether this decomposition is lossy or not.

For lossy decomposition, we must have-

$$R_1 \bowtie R_2 \supset R$$

Now, if we perform the natural join ($\bowtie$) of the sub relations R_1 and R_2 we get-

A	B	C
1	2	1
2	5	3
2	3	3
3	5	3
3	3	3

This relation is not same as the original relation R and contains some extraneous tuples.

Clearly, $R_1 \bowtie R_2 \supset R$.

Thus, we conclude that the above decomposition is lossy join decomposition.

NOTE-

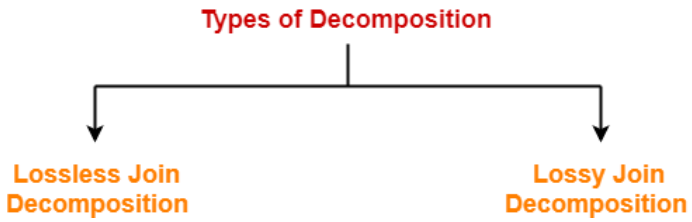
- Lossy join decomposition is also known as **careless decomposition.**
- This is because extraneous tuples get introduced in the natural join of the sub-relations.
- Extraneous tuples make the identification of the original tuples difficult.

Determine Decomposition Is Lossless Or Lossy

Decomposition in DBMS-

We have discussed-

- Decomposition is a process of dividing a single relation into two or more sub relations.
- Decomposition of a relation can be completed in the following two ways-



1. Lossless Join Decomposition
2. Lossy Join Decomposition

In this session, we will learn how to determine whether the decomposition is lossless or lossy.

Determining Whether Decomposition Is Lossless Or Lossy-

Consider a relation R is decomposed into two sub relations R_1 and R_2.

Then,

- If all the following conditions satisfy, then the decomposition is lossless.
- If any of these conditions fail, then the decomposition is lossy.

Condition-01:

Union of both the sub relations must contain all the attributes that are present in the original relation R.

Thus,

$$R_1 \cup R_2 = R$$

Condition-02:

- Intersection of both the sub relations must not be null.
- In other words, there must be some common attribute that is present in both the sub relations.

Thus,

$$R1 \cap R2 \neq \emptyset$$

Condition-03:

The intersection of both the sub relations must be a super key of either R_1 or R_2 or both.

Thus,

$$R_1 \cap R_2 = \text{Super key of } R_1 \text{ or } R_2$$

PRACTICE PROBLEMS BASED ON DETERMINING WHETHER DECOMPOSITION IS LOSSLESS OR LOSSY-

Problem-01:

Consider a relation schema R (A , B , C , D) with the functional dependencies A → B and C → D. Determine whether the decomposition of R into R_1 (A , B) and R_2 (C , D) is lossless or lossy.

Solution-

To determine whether the decomposition is lossless or lossy,

- We will check all the conditions one by one.
- If any of the conditions fail, then the decomposition is lossy otherwise lossless.

Condition-01:

According to condition-01, union of both the sub relations must contain all the attributes of relation R.

So, we have-

$$R_1 (A , B) \cup R_2 (C , D)$$

$$= R (A , B , C , D)$$

Clearly, union of the sub relations contain all the attributes of relation R.

Thus, condition-01 satisfies.

Condition-02:

According to condition-02, intersection of both the sub relations must not be null.

So, we have-

$$R_1 (A , B) \cap R_2 (C , D)$$

$$= \Phi$$

Clearly, intersection of the sub relations is null.

So, condition-02 fails.

Thus, we conclude that the decomposition is lossy.

Problem-02:

Consider a relation schema R (A , B , C , D) with the following functional dependencies-

$$A \rightarrow B$$

$$B \rightarrow C$$

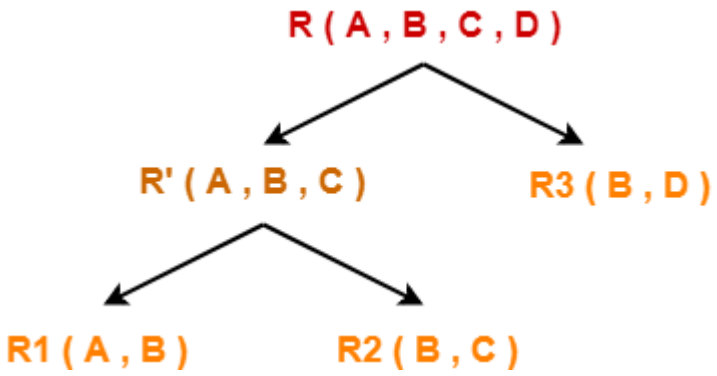
$$C \rightarrow D$$

$$D \rightarrow B$$

Determine whether the decomposition of R into R_1 (A , B) , R_2 (B , C) and R_3 (B , D) is lossless or lossy.

Solution-

Consider the original relation R was decomposed into the given sub relations as shown-



Decomposition of R(A, B, C, D) into R'(A, B, C) and R_3(B, D)-

To determine whether the decomposition is lossless or lossy,

- We will check all the conditions one by one.
- If any of the conditions fail, then the decomposition is lossy otherwise lossless.

Condition-01:

According to condition-01, union of both the sub relations must contain all the attributes of relation R.

So, we have-

$$R' (A , B , C) \cup R_3 (B , D)$$

$$= R (A , B , C , D)$$

Clearly, union of the sub relations contain all the attributes of relation R.

Thus, condition-01 satisfies.

Condition-02:

According to condition-02, intersection of both the sub relations must not be null.

So, we have-

$$R' (A , B , C) \cap R_3 (B , D)$$

$$= B$$

Clearly, intersection of the sub relations is not null.

Thus, condition-02 satisfies.

Condition-03:

According to condition-03, intersection of both the sub relations must be the super key of one of the two sub relations or both.

So, we have-

$$R' (A , B , C) \cap R_3 (B , D)$$

$$= B$$

Now, the closure of attribute B is-

$$B^+ = \{ B , C , D \}$$

Now, we see-

- Attribute ‘B’ can not determine attribute ‘A’ of sub relation R’.
- Thus, it is not a super key of the sub relation R’.
- Attribute ‘B’ can determine all the attributes of sub relation R_3.
- Thus, it is a super key of the sub relation R_3.

Clearly, intersection of the sub relations is a super key of one of the sub relations.

So, condition-03 satisfies.

Thus, we conclude that the decomposition is lossless.

Decomposition of R'(A, B, C) into R_1(A, B) and R_2(B, C)-

To determine whether the decomposition is lossless or lossy,

- We will check all the conditions one by one.
- If any of the conditions fail, then the decomposition is lossy otherwise lossless.

Condition-01:

According to condition-01, union of both the sub relations must contain all the attributes of relation R’.

So, we have-

$$R_1 (A , B) \cup R_2 (B , C)$$

$$= R' (A , B , C)$$

Clearly, union of the sub relations contain all the attributes of relation R’.

Thus, condition-01 satisfies.

Condition-02:

According to condition-02, intersection of both the sub relations must not be null.

So, we have-

$$R_1 (A , B) \cap R_2 (B , C)$$

$$= B$$

Clearly, intersection of the sub relations is not null.

Thus, condition-02 satisfies.

Condition-03:

According to condition-03, intersection of both the sub relations must be the super key of one of the two sub relations or both.

So, we have-

$$R_1 (A , B) \cap R_2 (B , C)$$

$$= B$$

Now, the closure of attribute B is-

$$B^+ = \{ B , C , D \}$$

Now, we see-

- Attribute ‘B’ can not determine attribute ‘A’ of sub relation R_1.
- Thus, it is not a super key of the sub relation R_1.
- Attribute ‘B’ can determine all the attributes of sub relation R_2.
- Thus, it is a super key of the sub relation R_2.

Clearly, intersection of the sub relations is a super key of one of the sub relations.

So, condition-03 satisfies.

Thus, we conclude that the decomposition is lossless.

Conclusion-

Overall decomposition of relation R into sub relations R_1, R_2 and R_3 is lossless.